Methods of proof – 2

Induction: strengthening, dimension, descent

Homework

13 сентября 2026 года

Theoretical minimum. Three reflexes. First: if the induction step does not close, strengthen the statement — replace a constant by a function of n, add an extra term, prove the inequality with a remainder. Second: choose the right parameter to induct on — dimension, rank, degree, number of prime factors, size of a block — and if two parameters are involved, order the pairs lexicographically. Third: over the integers a minimal counterexample always exists, so descent is available; over the reals it is not, and the substitute is compactness or a limiting argument. Zorn's lemma is the transfinite form of "take a maximal element" and is needed exactly when the object cannot be built in countably many steps.

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1. Prove that n lines in general position divide the plane into $\frac{n^2+n+2}{2}$ parts, and find and prove the analogous formula for hyperplanes in $\mathbb{R}^d$ by a double induction on n and d.

2. Prove by induction on the degree that a polynomial of degree n with $n+1$ integer values at consecutive integers has integer finite differences, and deduce that $\binom{x}{k}$ form a basis of the integer-valued polynomials (compare with Polynomials – 7).

3. Prove the Cauchy–Schwarz inequality by induction on the number of terms, and the AM–GM inequality by Cauchy's forward-backward induction; explain why the second one needs a non-standard induction scheme.

4. Prove that every finite group of order p^k has a non-trivial centre, by induction on k using the class equation, and deduce that a group of order p^2 is abelian.

5. Prove by descent that the equation $x^2+y^2+z^2 = 2xyz$ has no positive integer solutions, and that $x^3 + 2y^3 = 4z^3$ has none either.

6. Prove by induction on the rank that every matrix over a field is equivalent to a matrix with ones and zeros on the diagonal, and by induction on the size that a nilpotent matrix is similar to a strictly upper triangular one.

7. Show that the statement "every continuous function on $[0, 1]$ is bounded" becomes false if $[0, 1]$ is replaced by $[0, 1] \cap \mathbb{Q}$, so that no argument using only finitely many points at a time (such as an induction on a natural parameter) can prove it, and give the compactness proof; then explain the general rule about when an induction on $\mathbb{N}$ is available at all.

8. Prove by induction on the number of vertices that a planar graph has a vertex of degree at most five, and deduce the five-colour theorem.

9. Prove the existence of a maximal ideal in every ring with unit and of a maximal linearly independent subset in every vector space, each by Zorn's lemma; then show that for a countably generated space the countable procedure suffices.

10. Prove Sperner's lemma by induction on the dimension, and deduce Brouwer's fixed point theorem; compare with the one-dimensional case in Continuous functions – 5.